

Microcontroller-Based Intelligent Traffic Light System



ATISH GUPTA

Traffic light intervals are fixed independent of the traffic movement. So sometimes large red-light delays cause traffic congestion. Here is a microcontroller-based intelligent traffic light system that reduces the possibility of traffic jams caused by traffic light delays to an extent.

The block diagram of the microcontroller-based intelligent traffic light system is shown in Fig. 1. The system has two infrared (IR) module pairs comprising IR transmitters and receivers (TX1-RX1 and TX2-RX2). The IR transmitters are placed on the roadside facing the IR receivers that are placed on the opposite side of the road. The two IR module pairs are

interval lapses, the vehicle count is re-initialised to zero to calculate the traffic profile again in the next controlling interval. The firmware of the system also allows the user to change the traffic profiles, vehicle counts, traffic light delays for each profile and controlling interval.

Here the system allows control of only one traffic light. Multiple traffic lights in different areas can be controlled by a centralised system. Further development includes recording of traffic profiles for the whole day, which can be further analysed to determine traffic conditions in different areas of the city at a specific time. Based on the analysis, the centralised system can also update the traffic lights delays, profiles and controlling interval of any traffic light.

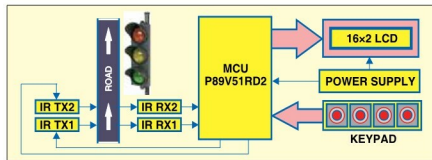


Fig. 1: Block diagram of the microcontroller-based intelligent traffic light system

PARTS LIST

Semiconductors:

IC1	- 7805 regulator
IC2	- P89V51RD2 microcontroller
T1-T5	- BC337 npn transistor
D1-D7	- 1N4007 rectifier diode
LED1	- 5mm LED
TX1, TX2	- IR LED
T6, T7	- L14F1 photo-transistor
LED	- 16x2-line LCD

Resistors (all 1/4-watt, $\pm 5\%$ per cent carbon)

R1	- 470-ohm
R2-R7	- 10-kilo-ohm
R8-R12	- 1-kilo-ohm
R13, R14	- 47-ohm
R15	- 100-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic
C3, C4	- 33pF ceramic
C5	- 10 μ F, 16V electrolytic

Miscellaneous:

X1	- 230V AC primary to 6V, 500mA secondary transformer
S1-S4	- Push-to-on tactile switch
X1	- 11.052MHz crystal
RL1-RL3	- 6V, 1C/O relays
B1	- 60W, 230V green bulb
B2	- 60W, 230V yellow bulb
B3	- 60W, 230V red bulb

placed about a metre apart so that the system can differentiate the vehicles from smaller-width objects like individuals.

The IR modules are situated far from traffic lights and activate when a vehicle passes the road between the transmitter and the receiver. The microcontroller controls the modules and counts the number of vehicles passing the road. Based on the vehicle count, the microcontroller firmware generates three traffic profiles—low, medium and high.

The microcontroller firmware contains different traffic light intervals (red, yellow and green light delays in seconds) for different traffic profiles. After calculating the traffic profile depending on the number of vehicles passing through the IR system in a pre-defined controlling interval (e.g., one minute), it further controls the timing of traffic lights. After the controlling

Circuit description

Fig. 2 shows the circuit of the microcontroller-based intelligent traffic light system. It comprises microcontroller P89V51RD2 (IC2), regulator IC 7805 (IC1), five BC337 transistors (T1 through T5), two L14F1 phototransistors (T6 and T7), an LCD module, and a few discrete components. Pin configurations of IC 7805 and transistor BC337 are shown in Fig. 3.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 6V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805. Capacitor C2 bypasses the ripples, if any, in the regulated supply. LED1 acts as the power-on indicator. Resistor R1 limits the current through LED1.

The microcontroller is the heart of the system. It is an 8-bit, 40-pin, low-power, high-performance device